# VU THANH DUC

Data Engineer | Python · SQL · PostgreSQL · AWS · Data Pipelines · LLM Integration
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## SUMMARY

Back-end engineer focused on data pipelines and data quality. Built a Python pipeline (OCR, LLM translation, TTS) with checkpoint/resume, retry with backoff, quality gates and model fallback (~23,700 cues translated, ~28,000 TTS segments). At FPT Software, built production .NET services on SQL Server: fixed stock race conditions with transactions and locking, and cut lookup latency from ~800ms to under 150ms with Redis. Led a team of 5 on a queue-based AWS SQS platform. Seeking a Data Engineer role.

## TECHNICAL SKILLS

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|---|---|
| **Languages** | Python, SQL, C#, TypeScript, JavaScript (ES2023+) |
| **Data Pipelines** | Python ETL pipelines, checkpoint/resume, retry/backoff and rate-limit handling, data-quality gates, queue-based workers (Amazon SQS), caching |
| **Databases** | PostgreSQL, SQL Server, MySQL, MongoDB, Redis; transactions and locking for data integrity |
| **Cloud & DevOps** | AWS (S3, SQS, RDS), DigitalOcean, Azure, Docker, Jenkins CI/CD, SonarQube |
| **AI / LLM** | OpenAI API, Google Gemini, Mistral/Cohere LLM APIs, Ollama (local LLMs), PaddleOCR, OpenCV, Faster Whisper, Pyannote.audio, Sentence Transformers, spaCy |
| **Backend** | .NET Core, Node.js (Express), RESTful APIs, Microservices, JWT/OAuth2 |
| **Frontend** | React.js, Next.js, Vue.js, TailwindCSS |
| **Tools** | Git/GitHub, Jira, xUnit/Moq, Agile/Scrum |

## PROJECTS

**Subtitle OCR, Translation & TTS Pipeline — data automation** *Jun 2026 – Present*
*Python · OpenCV · PaddleOCR (GPU) · Mistral / Cohere LLMs · Ollama · TTS*
*Personal project, used by friends; not publicly released*

- Built a two-stage pipeline: video to Chinese subtitles (OCR of burned-in text), then Vietnamese SRT plus a timeline-synced TTS mp3. Runs on a laptop GTX 1660 Ti at ~3.1x real time (≈19 min for a 1-hour video, OCR stage).
- **Data quality:** two-pass OCR (detect segments, then one OCR call per segment), consensus voting across frames to avoid fabricated lines, 10 fps recovery pass for missed cues. Result: 94.3% clean cues on a 14,996-cue file.
- **Reliability:** checkpoint/resume for long jobs, retry/backoff on rate limits, daily token budget, automatic fallback to local models (Ollama Qwen3, opus-mt).
- **Cost control:** Google Translate first pass; cloud LLMs refine only cues a scoring step marks weak, with a domain glossary and a quality gate. Scale to date: ~23,700 cues translated, ~28,000 TTS segments.

**OratorAI — AI Presentation Analysis Platform** *Jan – Apr 2026*
*Team Leader & Back-end Developer · Team of 5 · Academic project*

- Designed a microservices system for presentation analysis (Faster Whisper, Pyannote.audio, Sentence Transformers + spaCy); used Amazon SQS to decouple AI workers from the API for horizontal scaling.
- Integrated Google Gemini to auto-generate structured performance reports.

**Also (Team Leader, team of 5):** AImpact — AI personal finance app (Grok, PayOS) · Quizlingo — adaptive English quiz app (OpenAI) · EVCare — EV service management system (Gemini, PayOS)

## EXPERIENCE

**Back-end & Front-end Developer — FPT Software** *Jan 2025 – Jun 2026*

**Smart Inventory & Order Tracking System (APAC retail client)** *May 2025 – Oct 2025*
*.NET Core · SQL Server · Redis · Docker*

- Built APIs for order placement and real-time inventory sync; resolved low-stock race conditions with transactions and locking on SQL Server (no overselling in flash-sale trials).
- Added Redis caching, cutting stock-lookup response time from ~800ms to under 150ms; wrote xUnit/Moq tests covering 75%+ of key business logic.

**Internal Resource & Sprint Tracking Tool** *Jan 2025 – Apr 2025*
*Vue.js · Node.js (Express) · MongoDB · Docker*

- Built Express REST APIs and a Vue.js dashboard syncing Jira tasks and SonarQube coverage via webhooks; cut weekly report time by ~60%; used by 100+ people in the unit.

**Citizen Management Platform (Japan)** *Nov 2025 – Jun 2026*
*C# (.NET Core) · Vue.js · SQL Server · Jenkins CI/CD*

- Built and maintained C# (.NET) services for a government citizen management platform deployed in Japan; developed a shared Vue.js component library adopted across the product.

**Freelance Full-Stack Developer — Solo projects** *Jan 2026 – Sep 2026*

- **Happy Furniture** (happyfurniturevn.com): React storefront, Next.js admin, .NET REST API; shipped live from requirements to deployment, 30+ products.
- **Restaurant table-side ordering system:** QR ordering with real-time routing to a kitchen display (React, Node.js, MySQL, WebSocket), admin panel and daily sales reports.
- **I8 Studio** (i8.com.vn): live showcase site for a TVC production studio (Next.js, Prisma, Tailwind CSS v4).

## EDUCATION

**FPT University** — Bachelor of Software Engineering *Sep 2021 – May 2026*
**English:** Working proficiency — comfortable reading technical documentation.